



I2200: Digital Image processing

Lecture 2: Digital Image Fundamentals -- Sampling & Quantization



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The City University of New York (CUNY)

Some materials from Dr. Lexing Xie and Dr. Shahram Ebadollahi



Notice:

- **Our TA cannot access Brightspace**
- Please submit your HW and project to Mr. Ryan Goldberg (TA) by **email**:
- rgoldbe000@citymail.cuny.edu



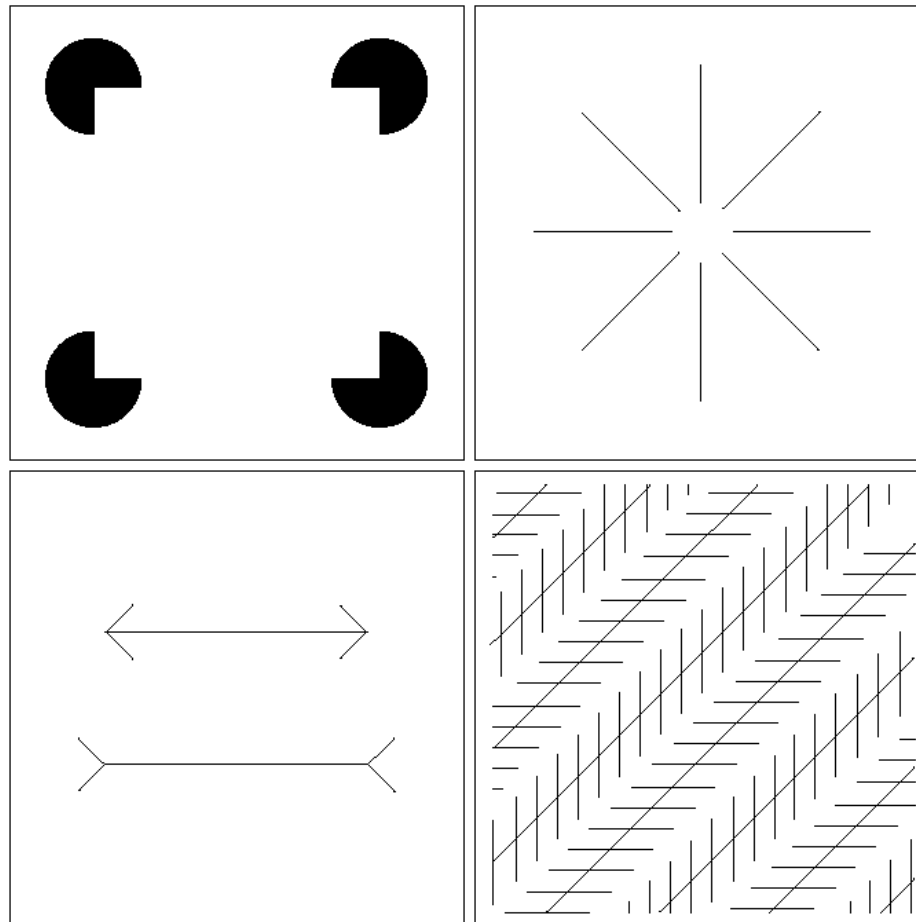
Outline:

- Illusions
- Image sampling and interpolation
- Image quantization
- Basic relationship between Pixels
- Basic operations for image processing
- Announcement:
 - HW1 is out today: Due: 2/17 (the day before class)
 - Read Chapter 2 this week

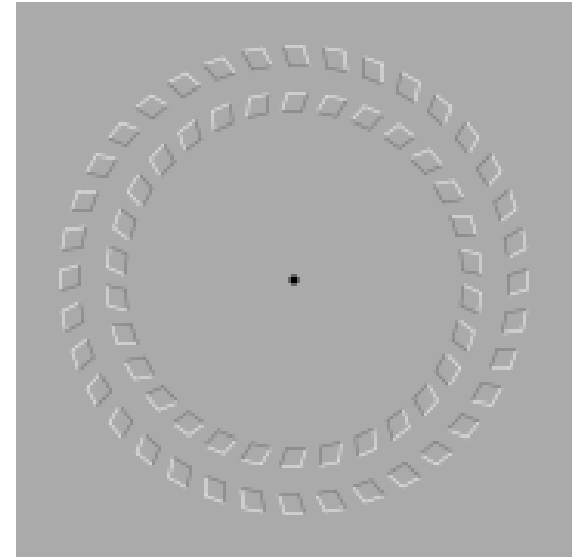
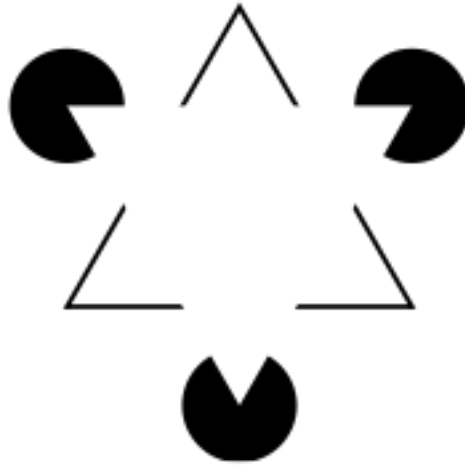
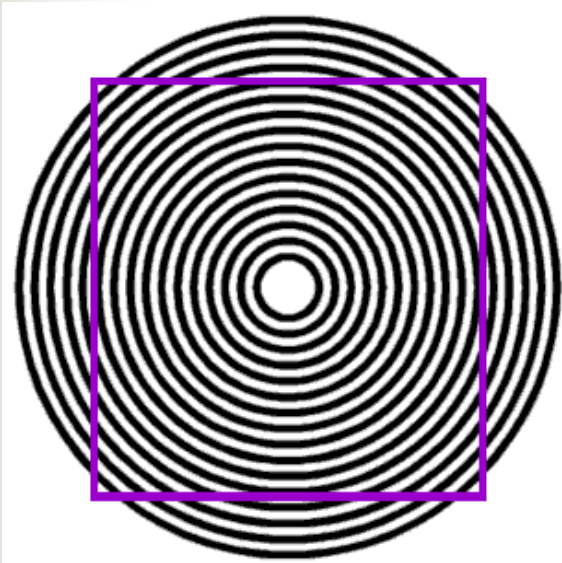
Cognitive illusions of the human visual system

a b
c d

FIGURE 2.9 Some well-known optical illusions.



Cognitive illusions of the human visual system

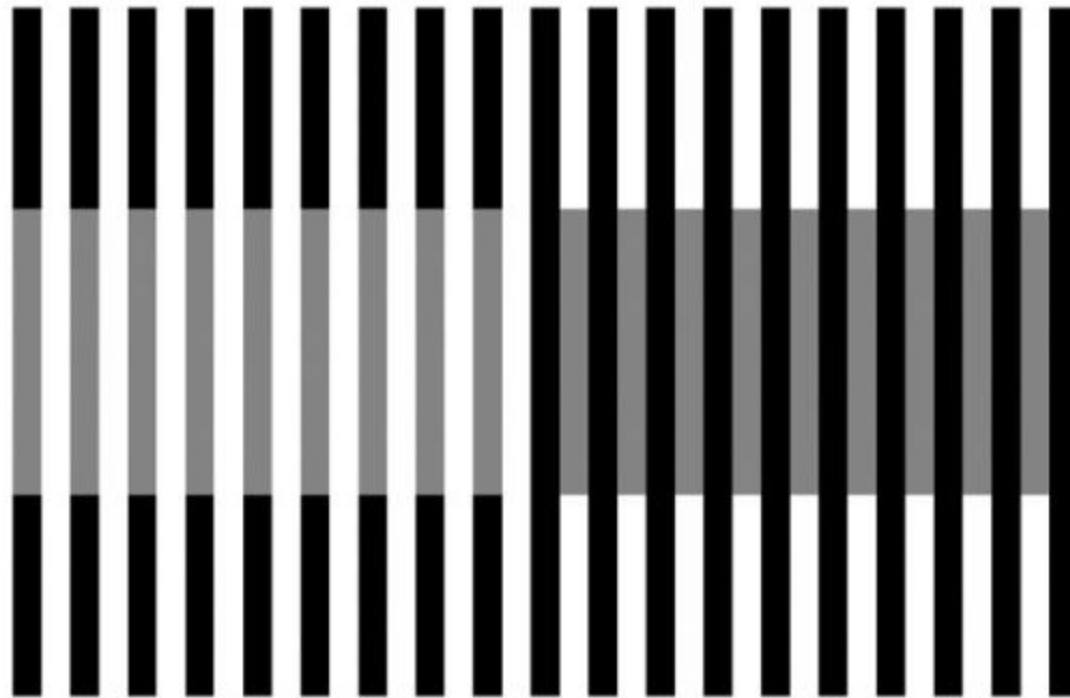


Cognitive illusions of the human visual system

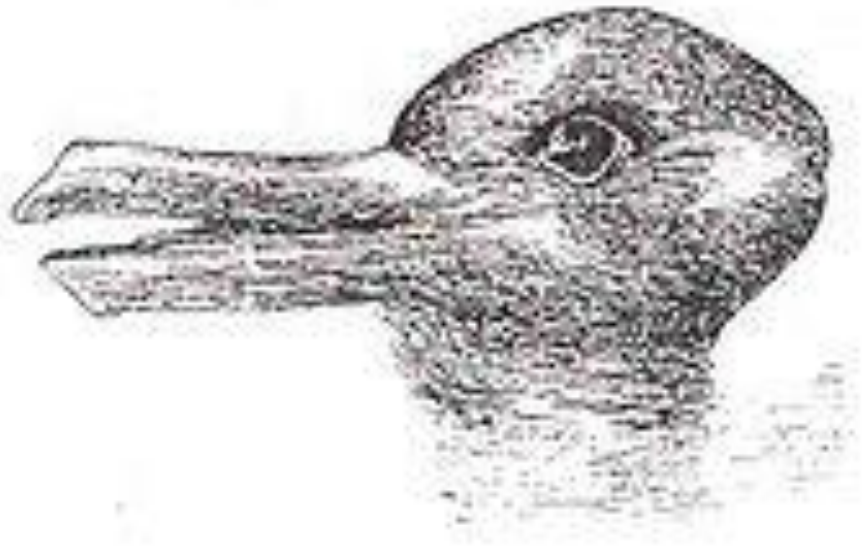
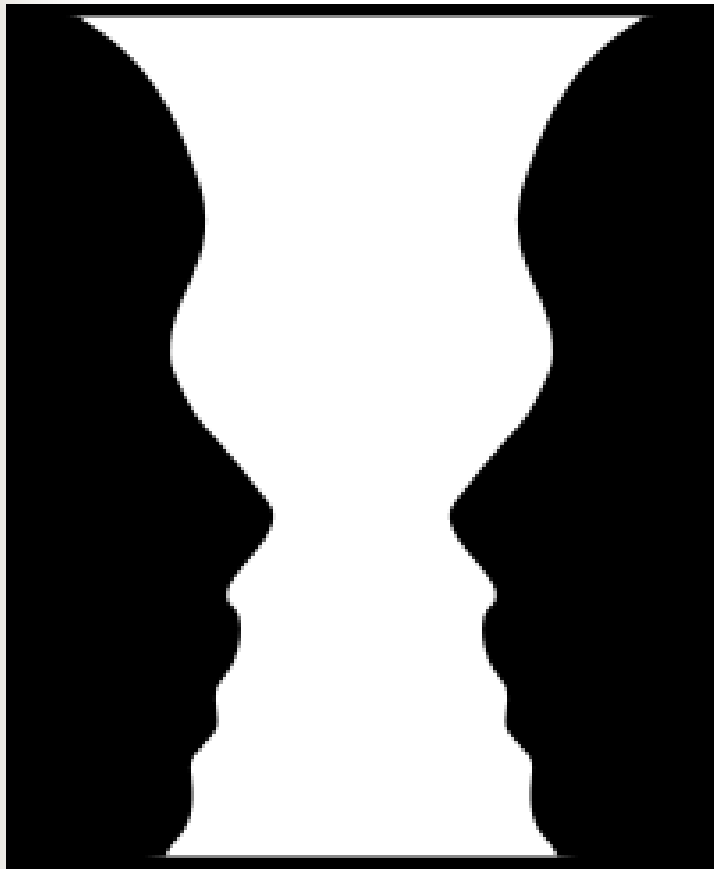


FIGURE 2.8 Examples of simultaneous contrast. All the inner squares have the same intensity, but they appear progressively darker as the background becomes lighter.

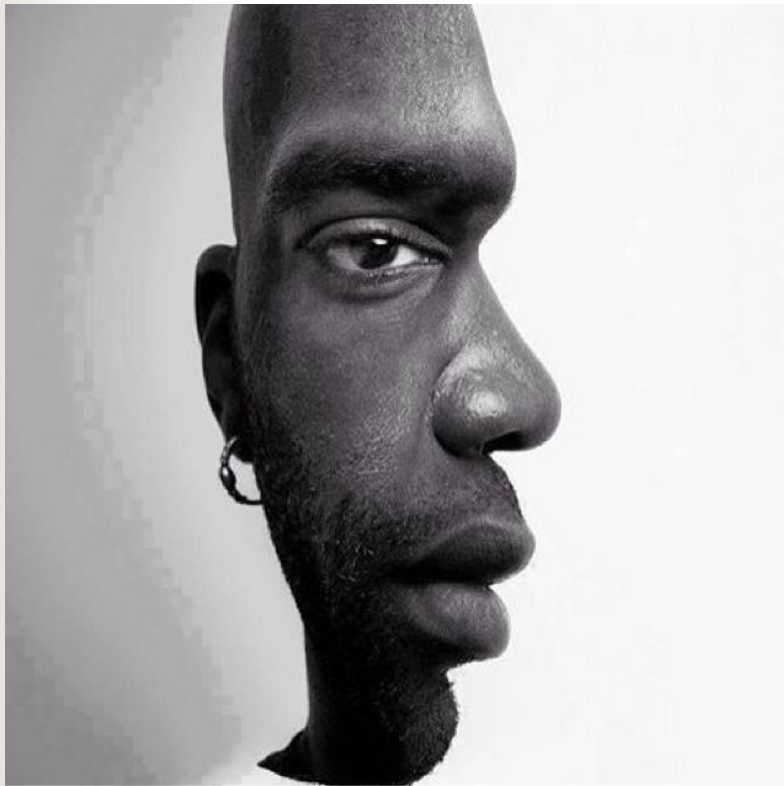
Cognitive illusions of the human visual system



Cognitive illusions of the human visual system



Cognitive illusions of the human visual system



<https://www.youtube.com/watch?v=7f1G6Nx5VDw>

Cognitive illusions of the human visual system





Confused color

red blue orange purple
orange blue green red
blue purple green red
orange blue red green
purple orange red blue
green red blue purple
orange blue red green
green purple orange red

Read out the color of
the words, not the
word itself.

How far can you go?

Digital Image Acquisition: From Physical Image to Digital Image

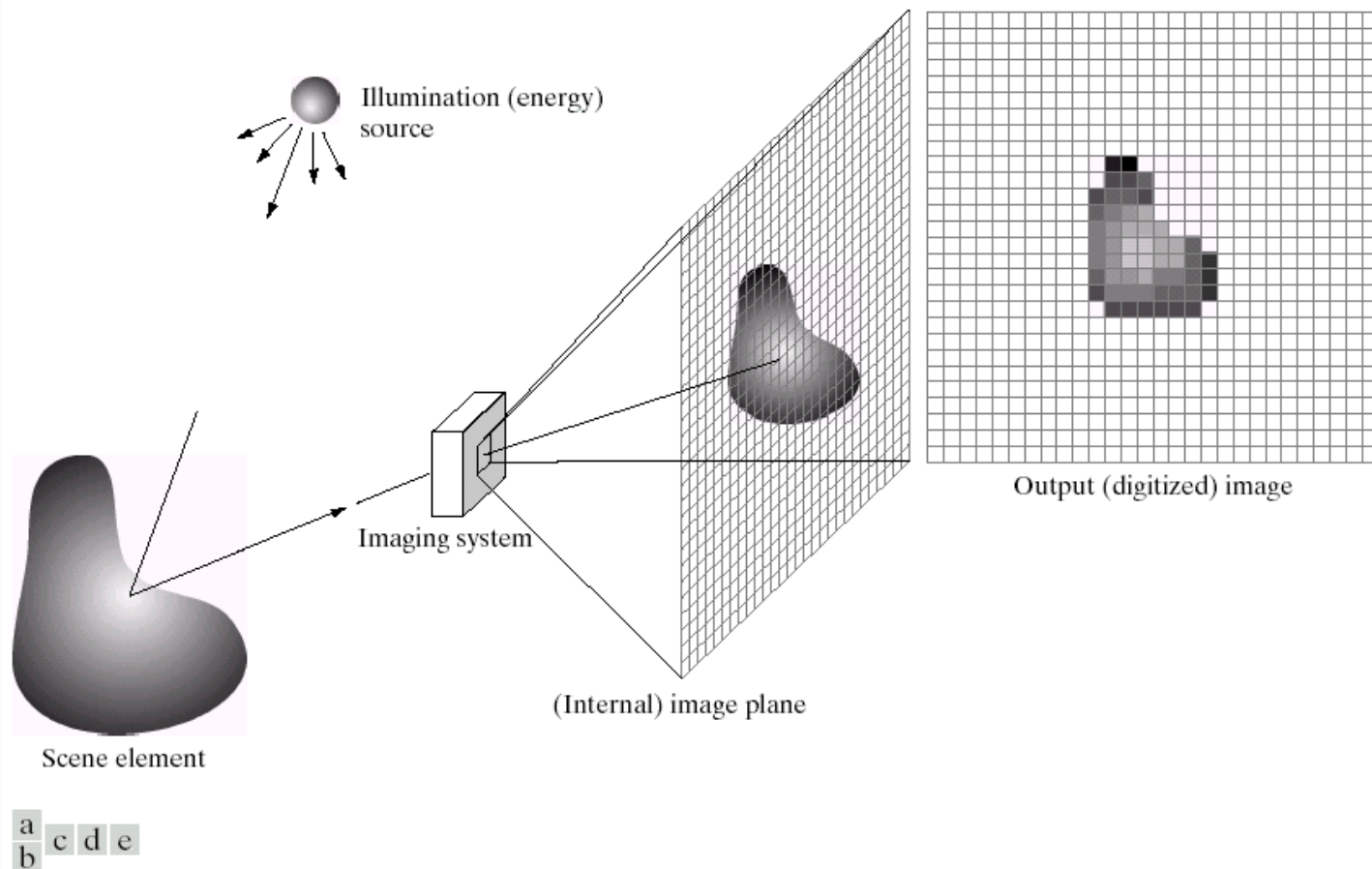
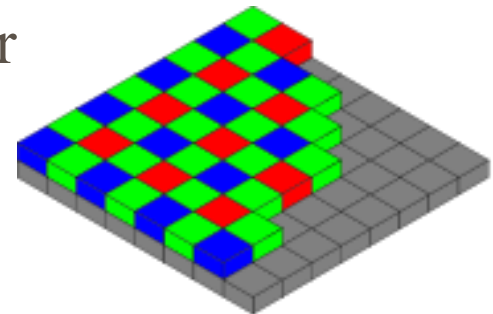


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy (“illumination”) source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

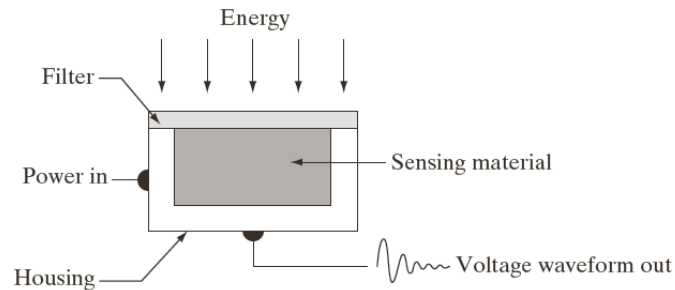
Digital Image Acquisition: From Physical Image to Digital Image

- Film
- Image sensors
 - CCD (Charge-coupled device)
 - http://en.wikipedia.org/wiki/CCD_camera
 - CMOS (Complementary Metal–Oxide–Semiconductor)
 - http://en.wikipedia.org/wiki/Active_pixel_sensor
 - Color Cameras: Bayer color filter



Digital Image Acquisition: CCD Sensor

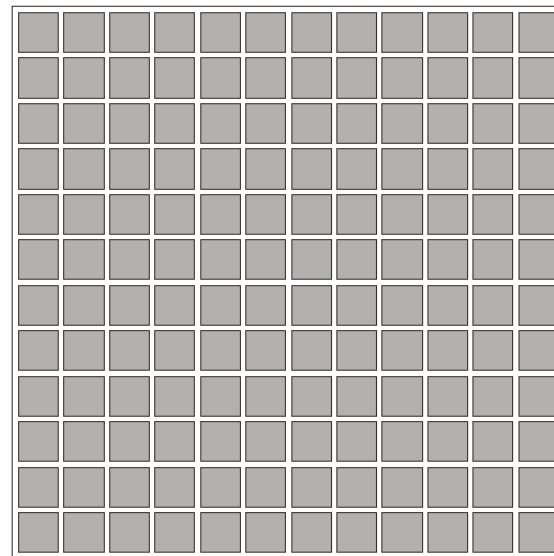
Single imaging sensor



Line sensor



Array sensor



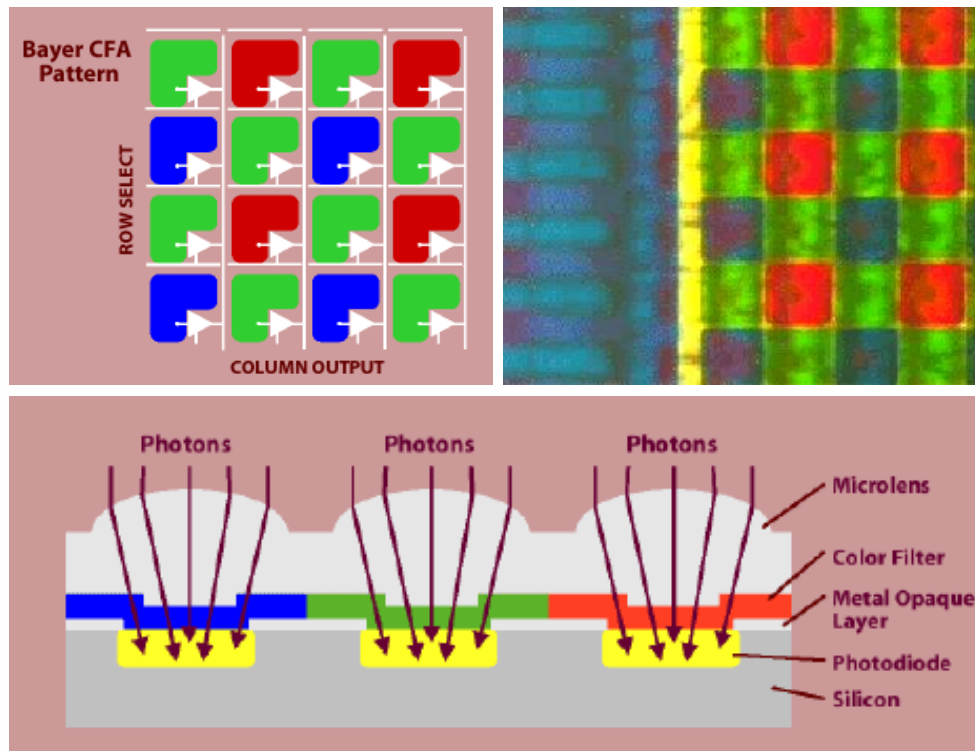


CCD vs CMOS

- A CCD is an analog device. When light strikes the chip it is held as a small electrical charge in each photo sensor. The charges are converted to voltage one pixel at a time as they are read from the chip. Additional circuitry in the camera converts the voltage into digital information. **More expensive, very low noise, slow.**
- A CMOS chip is a type of active pixel sensor made using the CMOS semiconductor process. Extra circuitry next to each photo sensor converts the light energy to a voltage. Additional circuitry on the chip may be included to convert the voltage to digital data. **Less expensive, low noise, fast.**

Bayer color filter

- Invented by B. E. Bayer at Kodak,
For more details:
<http://www.siliconimaging.com/RGB%20Bayer.html>





Digital Image Acquisition: From Physical Image to Digital Image

Goal: generate digital images from sensed data

■ Sampling

Process of mapping a continuous function to discrete – coordinate values

NOTE: pixels are samples from physical image

- Photoreceptors in eye
- CCD array

■ Quantization

Process of mapping continuous variable to discrete – digitizing the amplitude values

Digital Image Acquisition: From Physical Image to Digital Image



Continuous image



Digital image



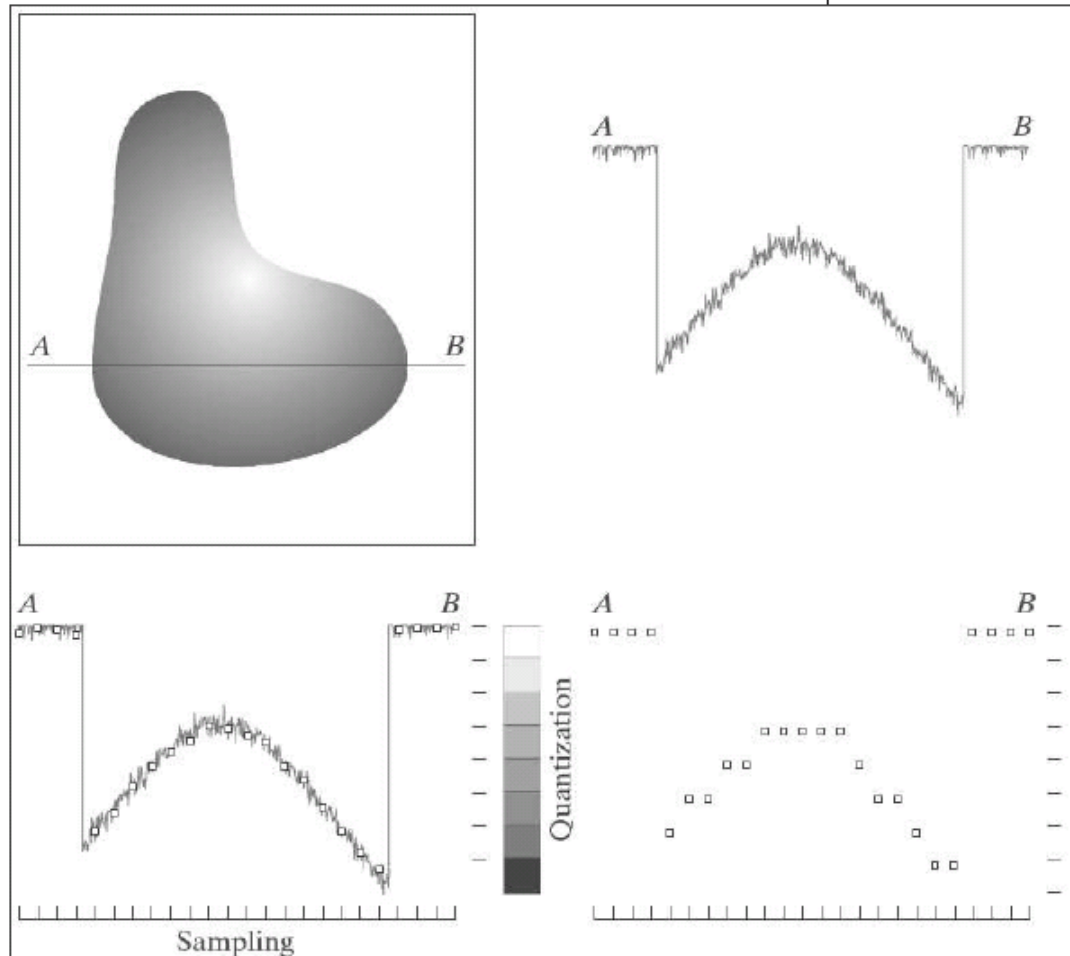
Image Resolution

- Intensity resolution
 - Each pixel has only “Depth” bits for colors/intensities
- Spatial resolution
 - Image has only “Width” x “Height” pixels
- Temporal resolution
 - Monitor refreshes images at only “Rate” Hz

Typical
Resolutions

	<u>Width x Height</u>	<u>Depth</u>	<u>Rate</u>
NTSC	640 x 480	8	30
Workstation	1280 x 1024	24	75
Film	3000 x 2000	12	24
Laser Printer	6600 x 5100	1	-

Digital Image Acquisition: From Physical Image to Digital Image



Spatial
discretization

Intensity
discretization

Digital Image Acquisition: From Physical Image to Digital Image

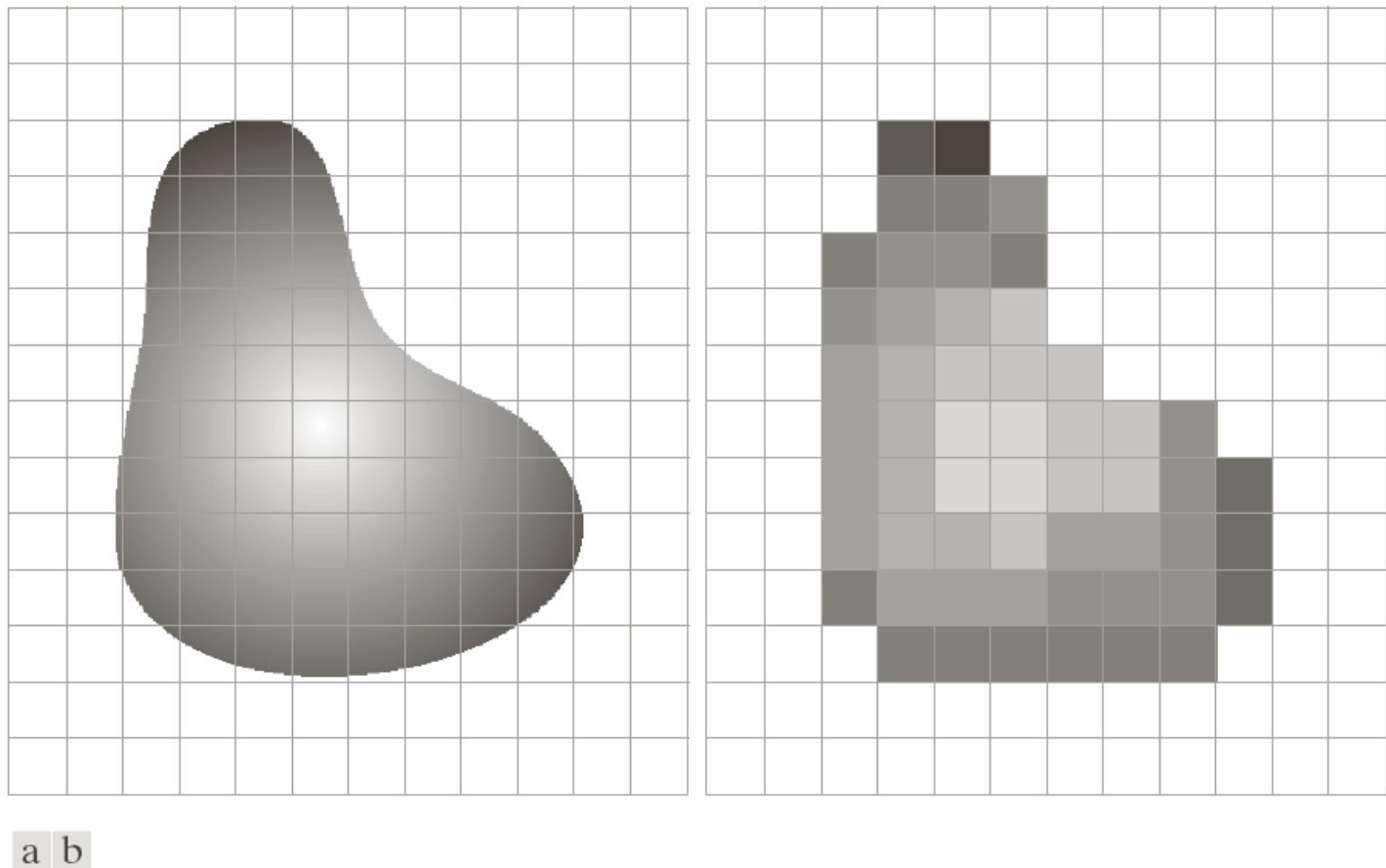
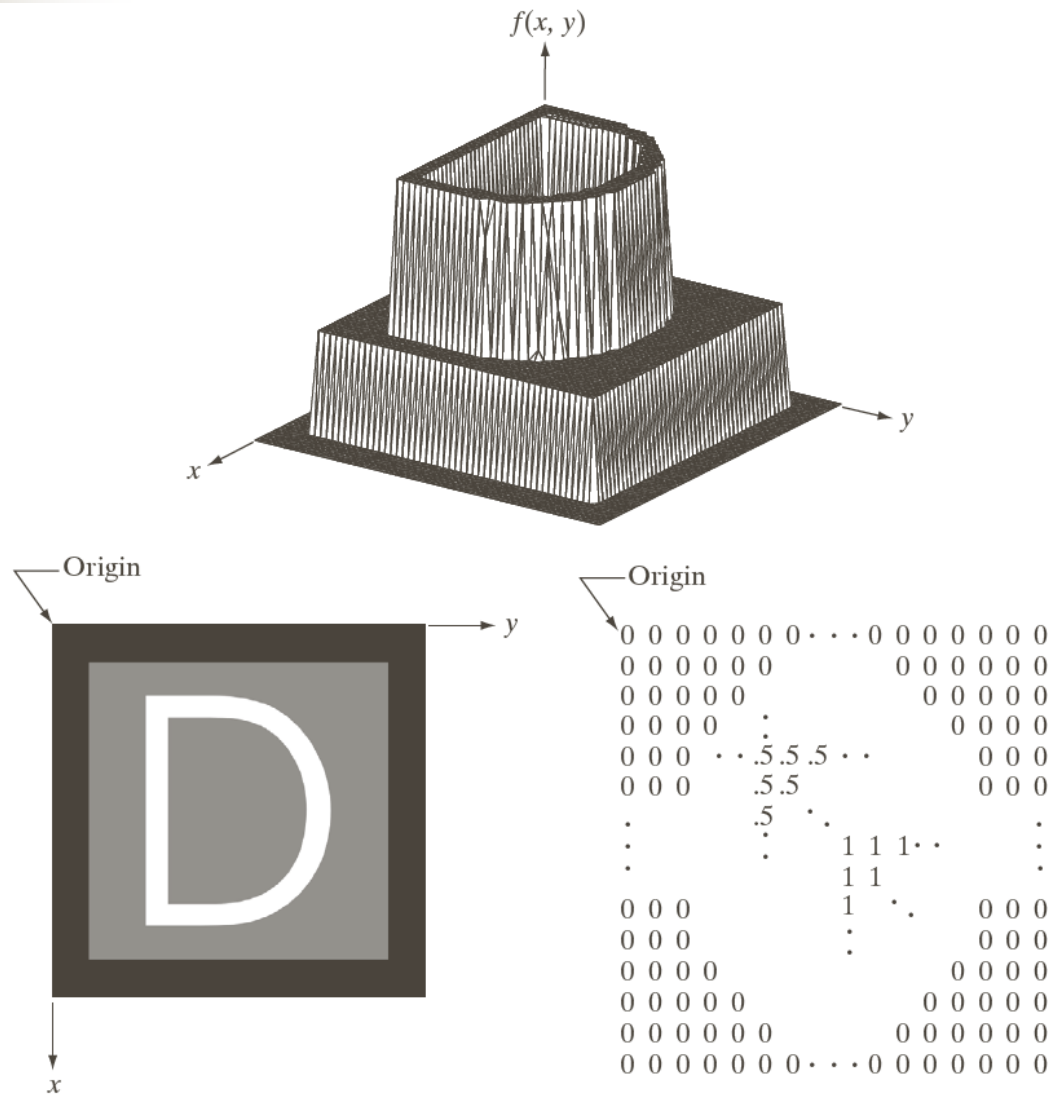


FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

Image representation



a
b c

FIGURE 2.18

(a) Image plotted as a surface.

(b) Image displayed as a visual intensity array.

(c) Image shown as a 2-D numerical array (0, .5, and 1 represent black, gray, and white, respectively).

Image representation - Array

$$f = \begin{bmatrix} f(0,0) & f(0,1) & \cdots & f(0,N-1) \\ f(1,0) & f(1,1) & \cdots & f(1,N-1) \\ \vdots & \vdots & \vdots & \vdots \\ f(M-1,0) & f(M-1,1) & \cdots & f(M-1,N-1) \end{bmatrix}$$

$$0 \leq f \leq L-1$$

$$0 \leq x \leq M-1 \quad 0 \leq y \leq N-1$$



Image Spatial and Intensity Resolution

Spatial resolution:

$$0 \leq x \leq M-1$$

$$0 \leq y \leq N-1$$

Note: change spatial resolution is called **sampling**.

Intensity resolution:

$$0 \leq f \leq L-1 \quad L \text{ is the number of intensity levels}$$

$$L = 2^k$$

Note: change intensity resolution is called **quantization**.



Image Storage

The number, b , of bits required to store a digitized M by N Image (k is the bits of image):

$$b = M \times N \times k$$

When $M = N$,

$$b = N^2 k$$

Question: 1 byte = ? bits

1 byte = 8 bits

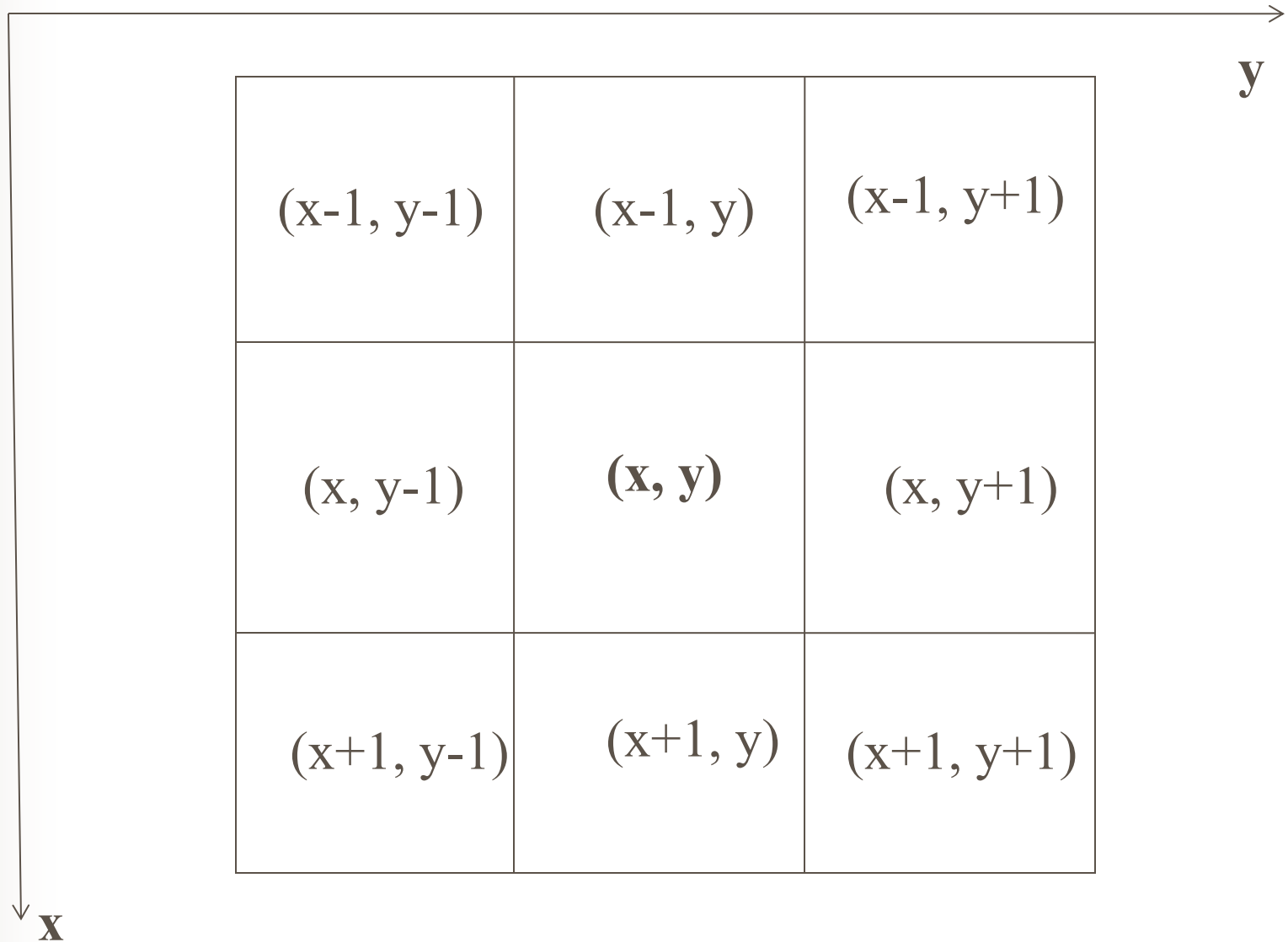
Image Storage

TABLE 2.1

Number of storage bits for various values of N and k .

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Neighbors of a pixel



Examples of Image of Different Spatial Resolution



Examples of Image of Different Intensity Resolution



1-bit



2-bit



3-bit



4-bit



5-bit



6-bit



7-bit



8-bit



Image Spatial Sampling --resizing

- High resolution → low resolution
 - Down-sample or shrink
- Low resolution → high resolution
 - Interpolation or zoom in

Image Interpolation

Interpolation works by using known data to estimate values at unknown points.

■ Linear interpolation

For example: if you wanted to know the temperature at noon, but only measured it at 11AM and 1PM,

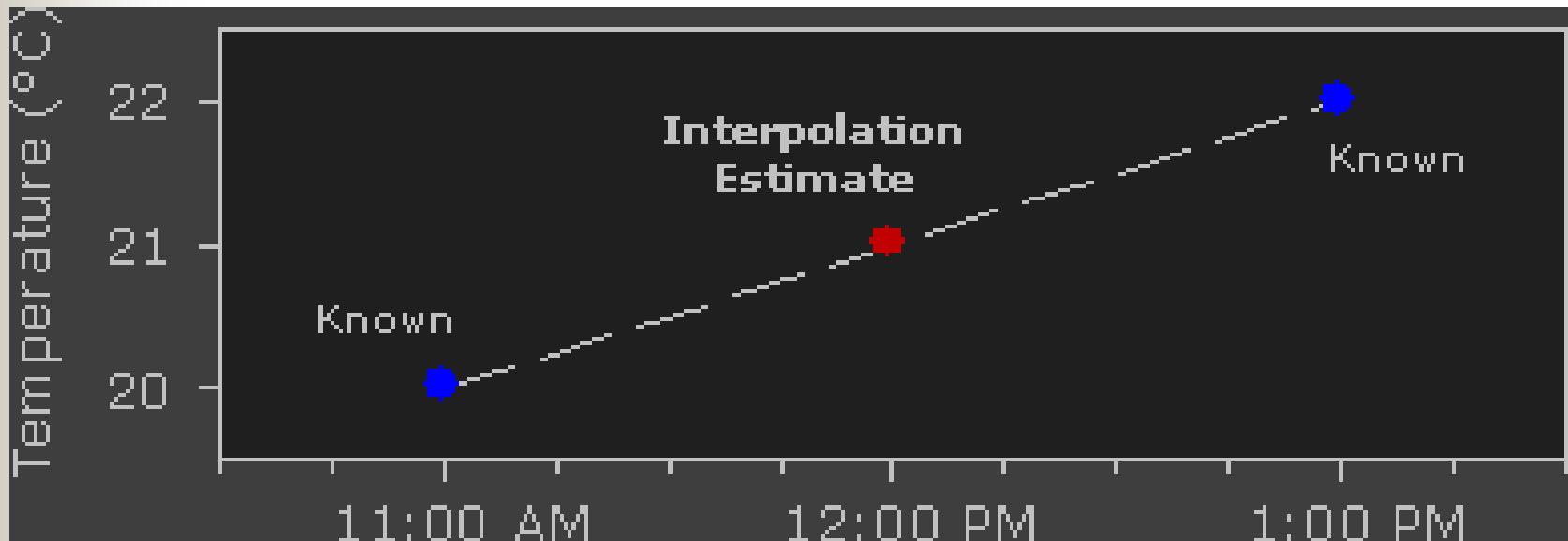


Image Interpolation

If you had an additional measurement at 11:30AM, you could see that the bulk of the temperature rise occurred before noon, and could use this additional data point to perform a quadratic interpolation:

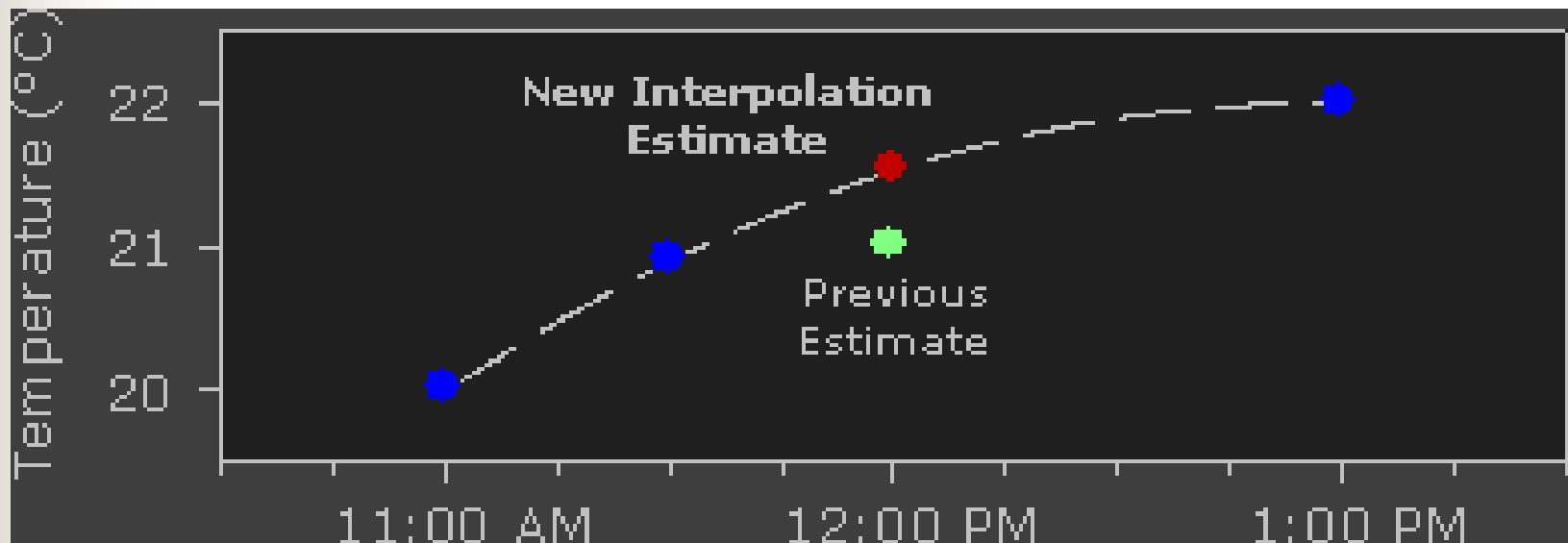
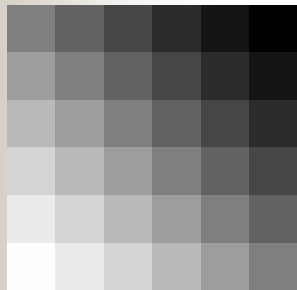


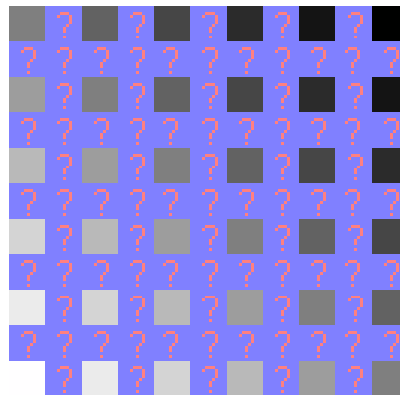
Image Interpolation

Image interpolation works in two directions and tries to achieve a best approximation of a pixel's color and intensity based on the values at surrounding pixels.

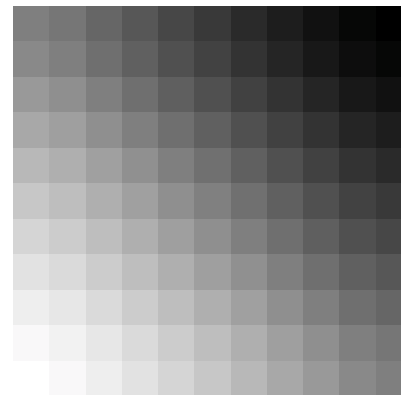
2D Interpolation 



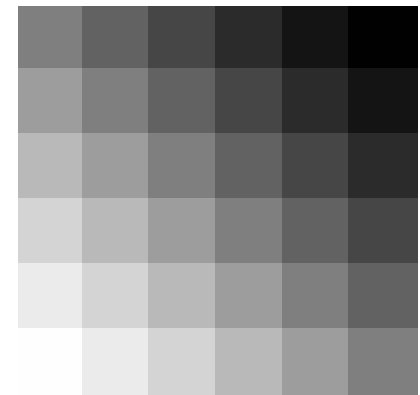
Original



Before



After



No Interpolation



Image Resizing methods

- **Non-adaptive algorithms**

- Nearest Neighbor Interpolation
- Bilinear interpolation
- Bicubic interpolation

- **Adaptive algorithms**

- Commercial software

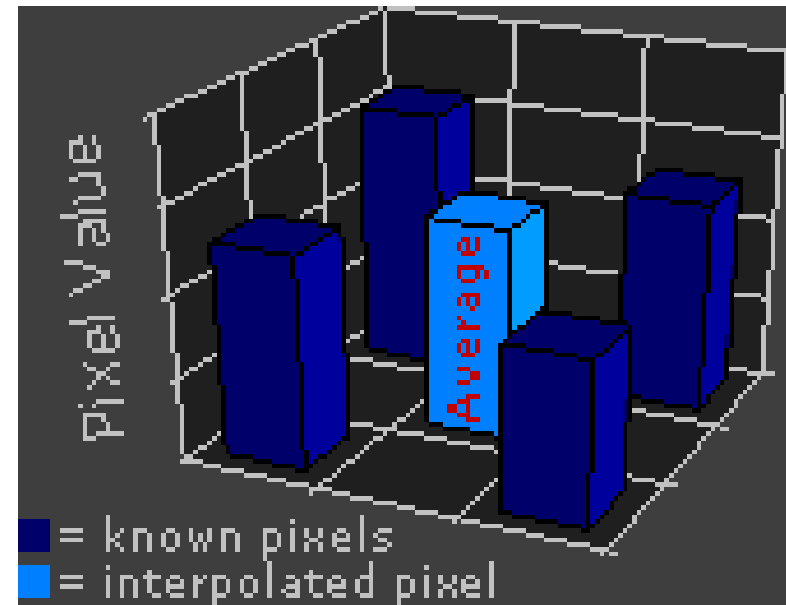


Nearest Neighbor Interpolation

- The most basic interpolation method
- Requires the least processing time of all the interpolation algorithms
- It only considers one pixel-- the closest one to the interpolated point.

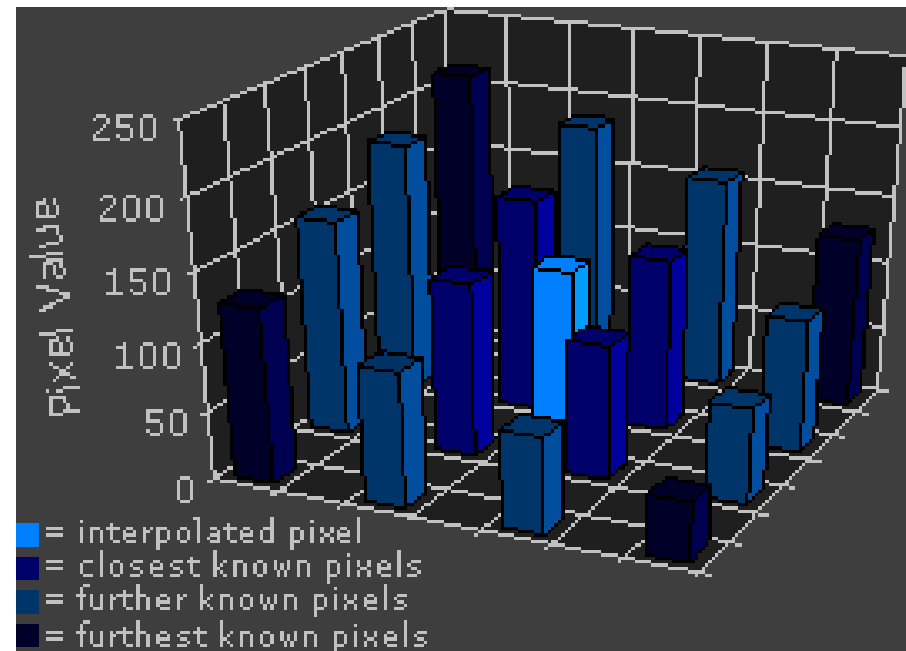
Bilinear Interpolation

- It considers the closest 2x2 neighborhood of known pixel values surrounding the unknown pixel.
- It then takes a weighted average of these 4 pixels to arrive at its final interpolated value.
- It results in much smoother looking images than nearest neighbor.

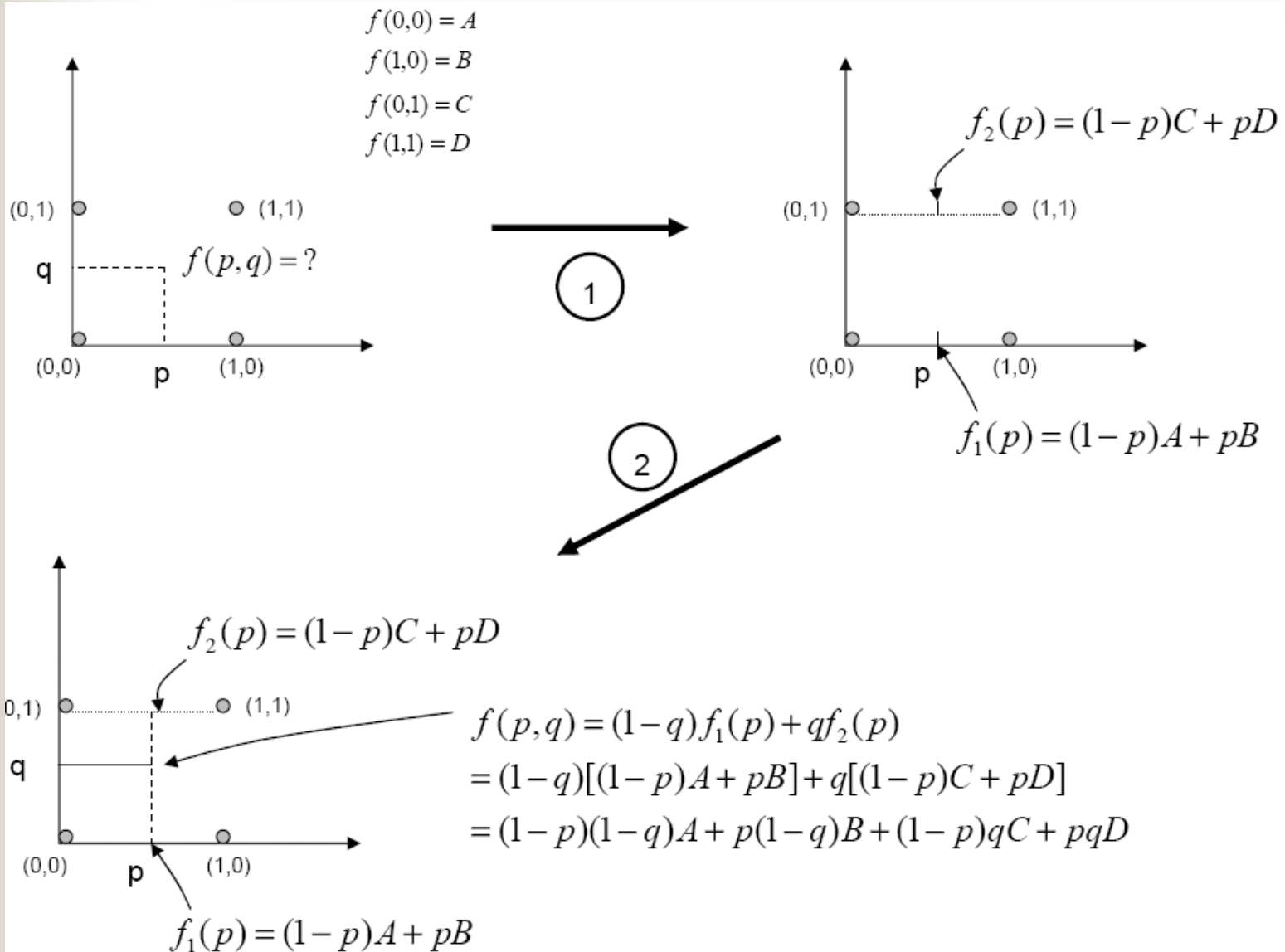


Bicubic Interpolation

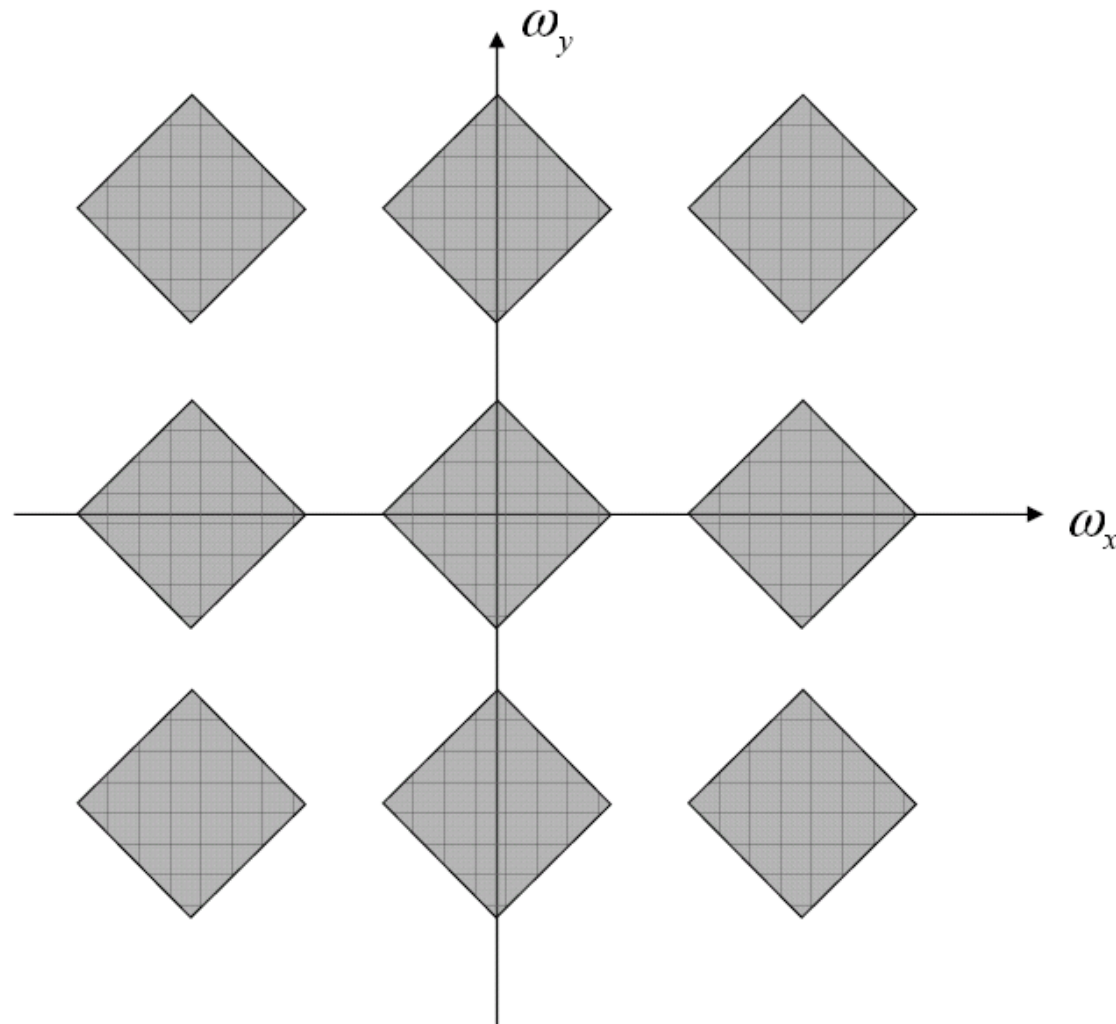
- It goes one step beyond bilinear by considering the closest 4x4 neighborhood of known pixels-- for a total of 16 pixels.
- Since these are at various distances from the unknown pixel, closer pixels are given a higher weighting in the calculation.



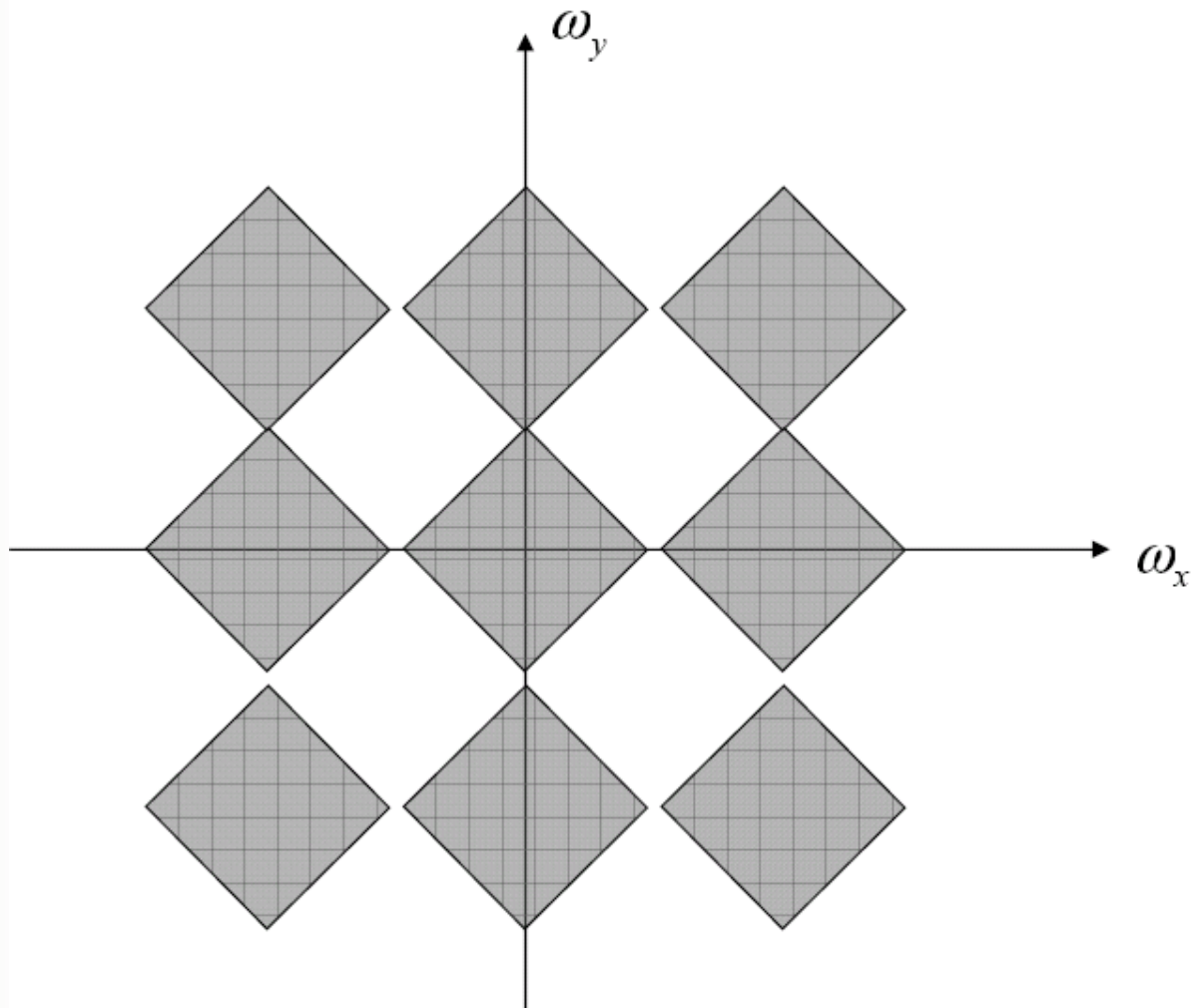
Bilinear Interpolation Method



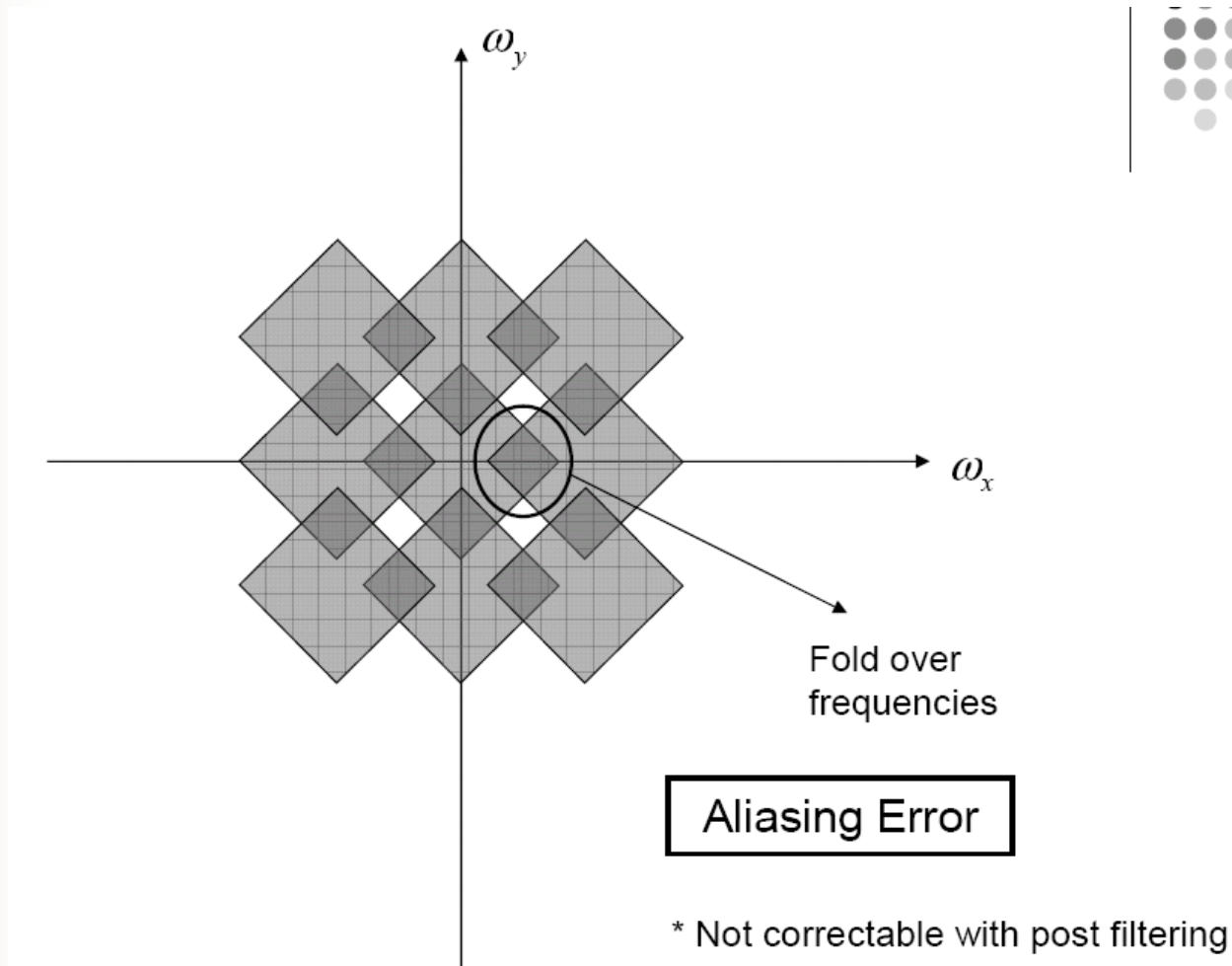
Problem with Downsampling



Problem with Downsampling



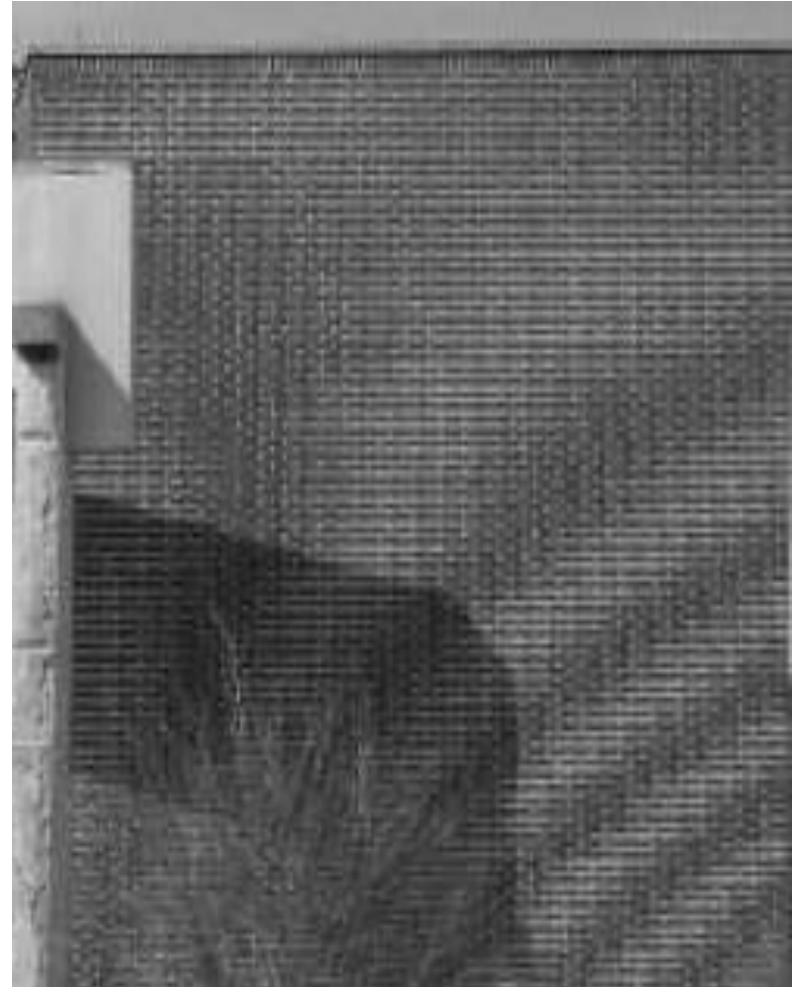
Problem with Downsampling



Aliasing Error – Visual Perception



Original image



Down-sampled
version



OPTICAL vs. DIGITAL ZOOM

Many compact digital cameras can perform both an optical and a digital zoom.

Optical zoom by moving the zoom lens so that it increases the magnification of light before it even reaches the digital sensor.

Digital zoom degrades quality by simply interpolating the image-- after it has been acquired at the sensor.

OPTICAL vs. DIGITAL ZOOM



10X Optical Zoom



10X Digital Zoom





5 mins break!

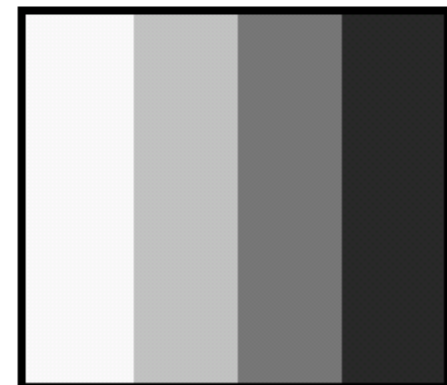
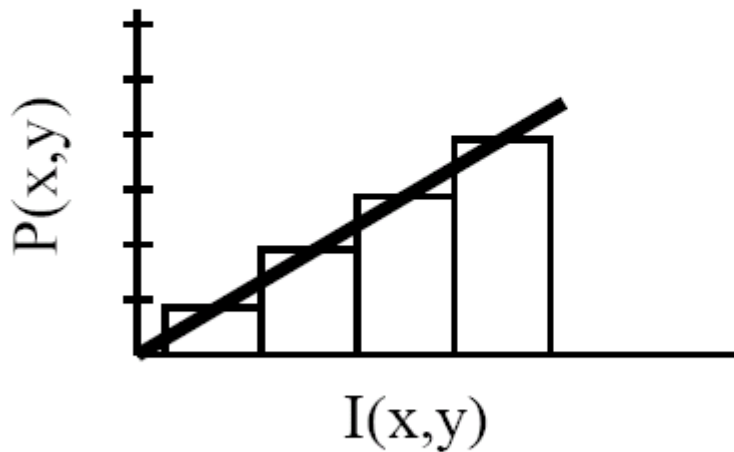


Quantization

- Artifacts due to limited intensity resolution
 - Frame buffers have limited number of bits per pixel
 - Physical devices have limited dynamic range

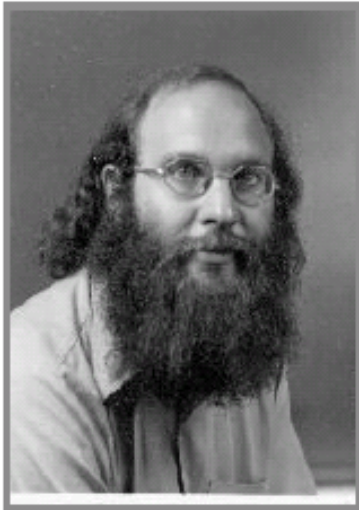
Uniform Quantization

$$P(x, y) = \text{trunc}(I(x, y) + 0.5)$$



Uniform Quantization

Images with decreasing bits per pixel:



8 bits



4 bits



2 bits



1 bit

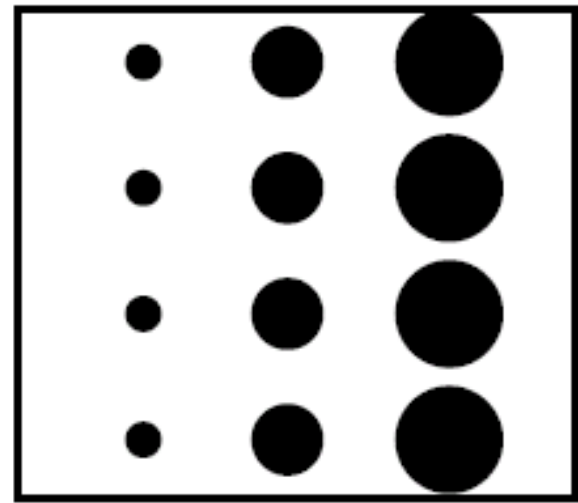
Reduce effects of quantization: Halftoning and Dithering

Classical Halftoning

- Use dots of varying size to represent intensities
 - Area of dots proportional to intensity in image

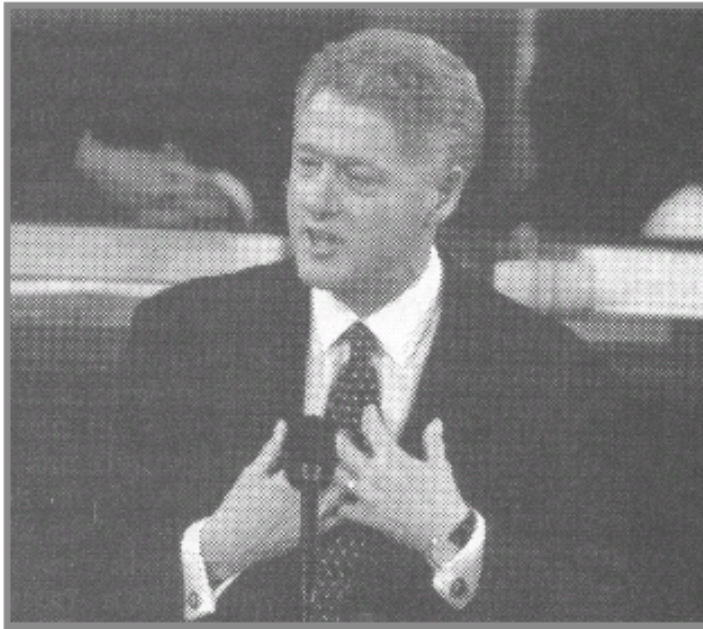


$I(x,y)$

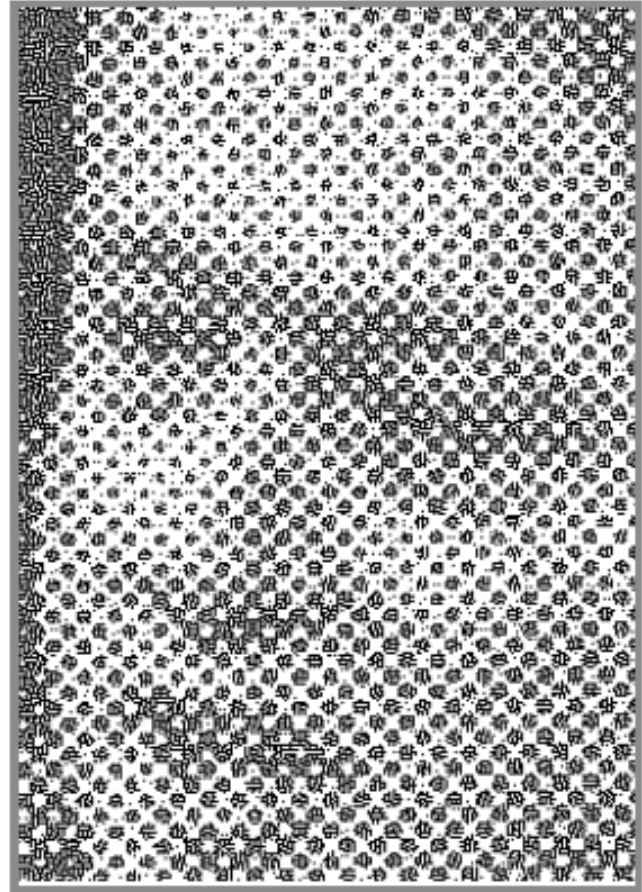


$P(x,y)$

Classical Halftoning

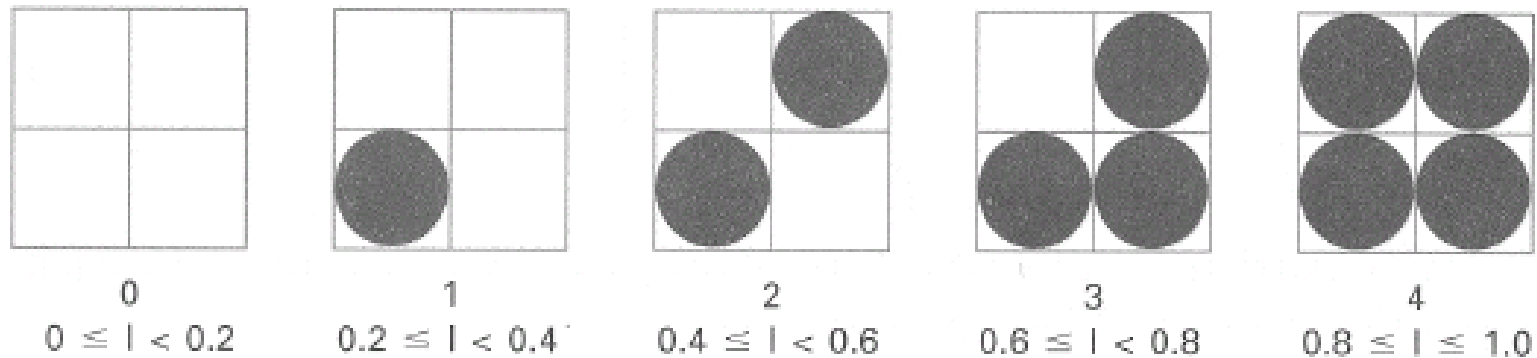


Newspaper Image



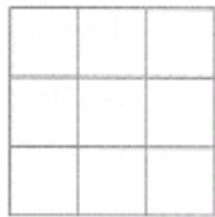
Halftone patterns

- Use cluster of pixels to represent intensity
 - Trade spatial resolution for intensity resolution

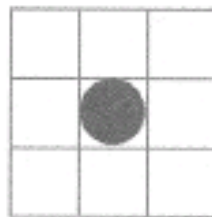


Halftone patterns

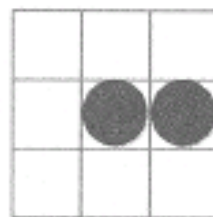
- How many intensities in a $n \times n$ cluster?



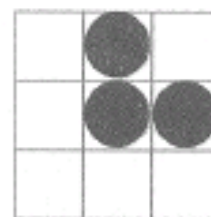
0
 $0 \leq I < 0.1$



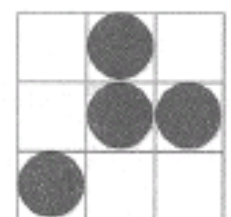
1
 $0.1 \leq I < 0.2$



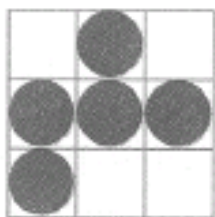
2
 $0.2 \leq I < 0.3$



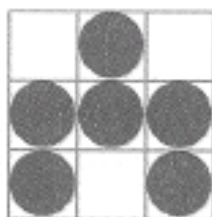
3
 $0.3 \leq I < 0.4$



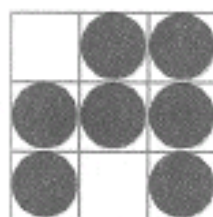
4
 $0.4 \leq I < 0.5$



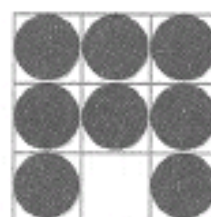
5
 $0.5 \leq I < 0.6$



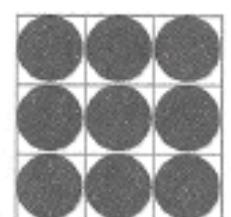
6
 $0.6 \leq I < 0.7$



7
 $0.7 \leq I < 0.8$



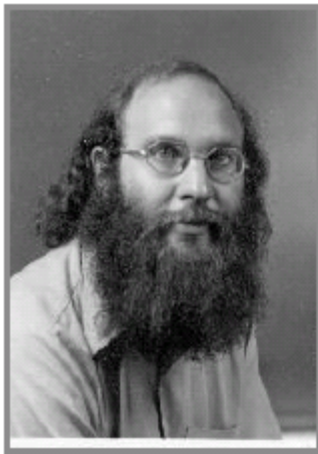
8
 $0.8 \leq I < 0.9$



9
 $0.9 \leq I \leq 1.0$

Dithering

- Distribute errors among pixels
 - Exploit spatial integration in our eye
 - Display greater range of perceptible intensities



Original
(8 bits)

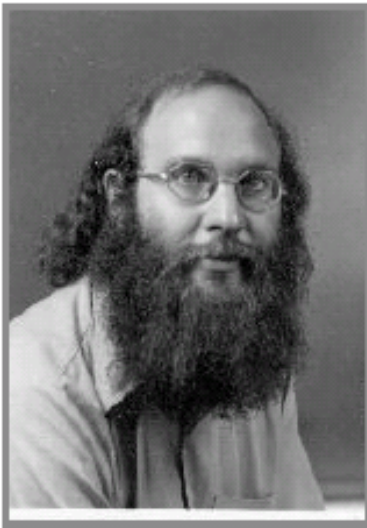


Uniform
Quantization
(1 bit)



Floyd-Steinberg
Dither
(1 bit)

Dithering: Random vs Ordered



Original
(8 bits)

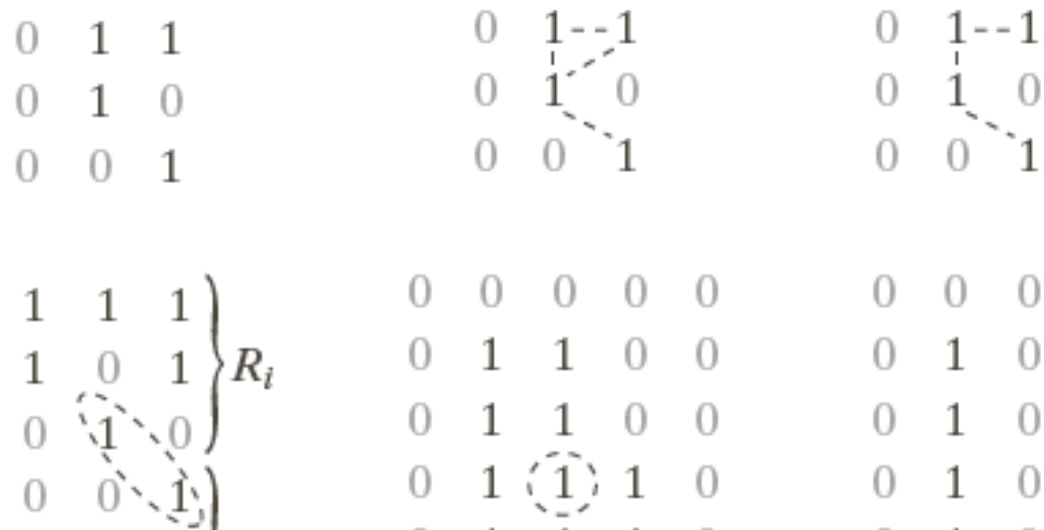


Random
Dither
(1 bit)



Ordered
Dither
(1 bit)

Relationship between pixels



a	b	c
d	e	f

FIGURE 2.25 (a) An arrangement of pixels. (b) Pixels that are 8-adjacent (adjacency is shown by dashed lines; note the ambiguity). (c) m -adjacency. (d) Two regions that are adjacent if 8-adjacency is used. (e) The circled point is part of the boundary of the 1-valued pixels only if 8-adjacency between the region and background is used. (f) The inner boundary of the 1-valued region does not form a closed path, but its outer boundary does.

Edge



Original image

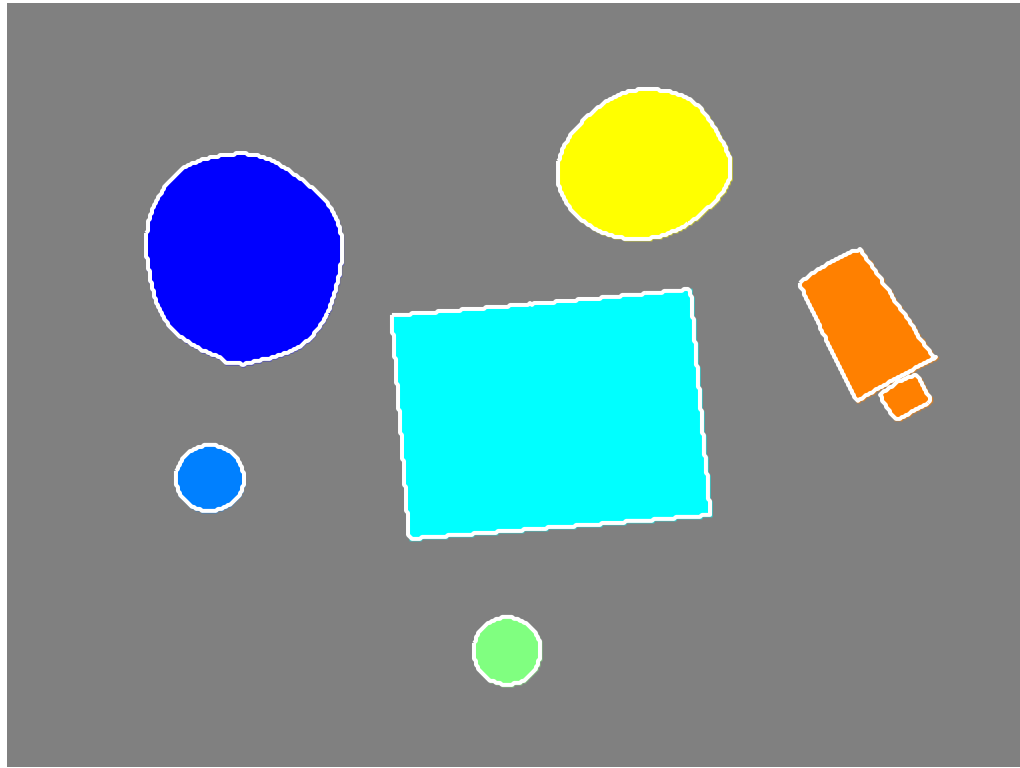


Sobel edge



Laplace edge

Boundary (border or contour)



Connected Component

- Connected components labeling scans an image and groups its pixels into components based on pixel connectivity



Connected Components Labeling



Distance Measurement

Pixels $p(x, y)$ and $q(s, t)$

- *Euclidean distance between p and q*

$$D(p, q) = \text{sqrt}((x-s)^2 + (y-t)^2)$$

- *D4 distance*

- $D4(p, q) = |x - s| + |y - t|$

- *D8 distance*

- $D8(p, q) = \max(|x - s|, |y - t|)$

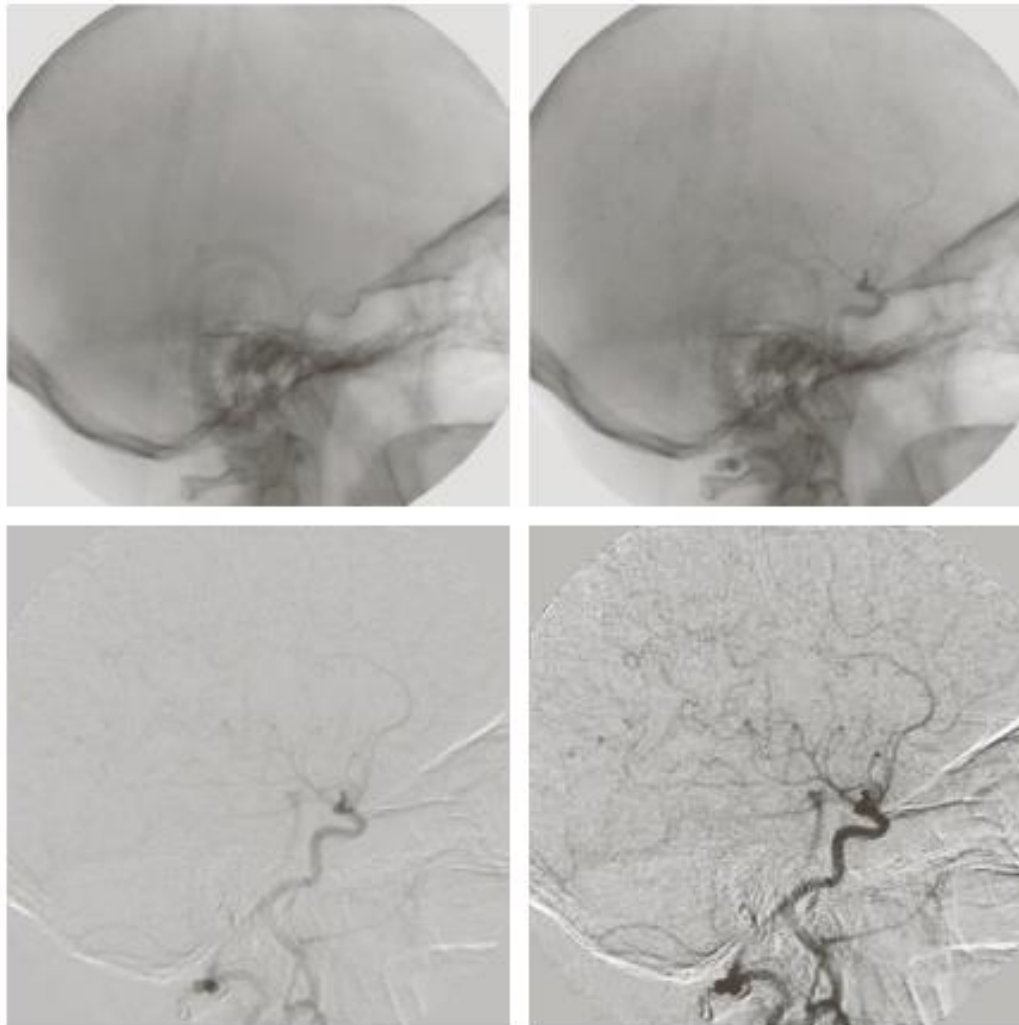
Image Difference



a b c

FIGURE 2.27 (a) Infrared image of the Washington, D.C. area. (b) Image obtained by setting to zero the least significant bit of every pixel in (a). (c) Difference of the two images, scaled to the range $[0, 255]$ for clarity.

Image difference and enhancement



a	b
c	d

FIGURE 2.28

Digital subtraction angiography.

(a) Mask image.

(b) A live image.

(c) Difference

between (a) and

(b). (d) Enhanced

difference image.

(Figures (a) and

(b) courtesy of

The Image

Sciences Institute,

University

Medical Center,

Utrecht, The

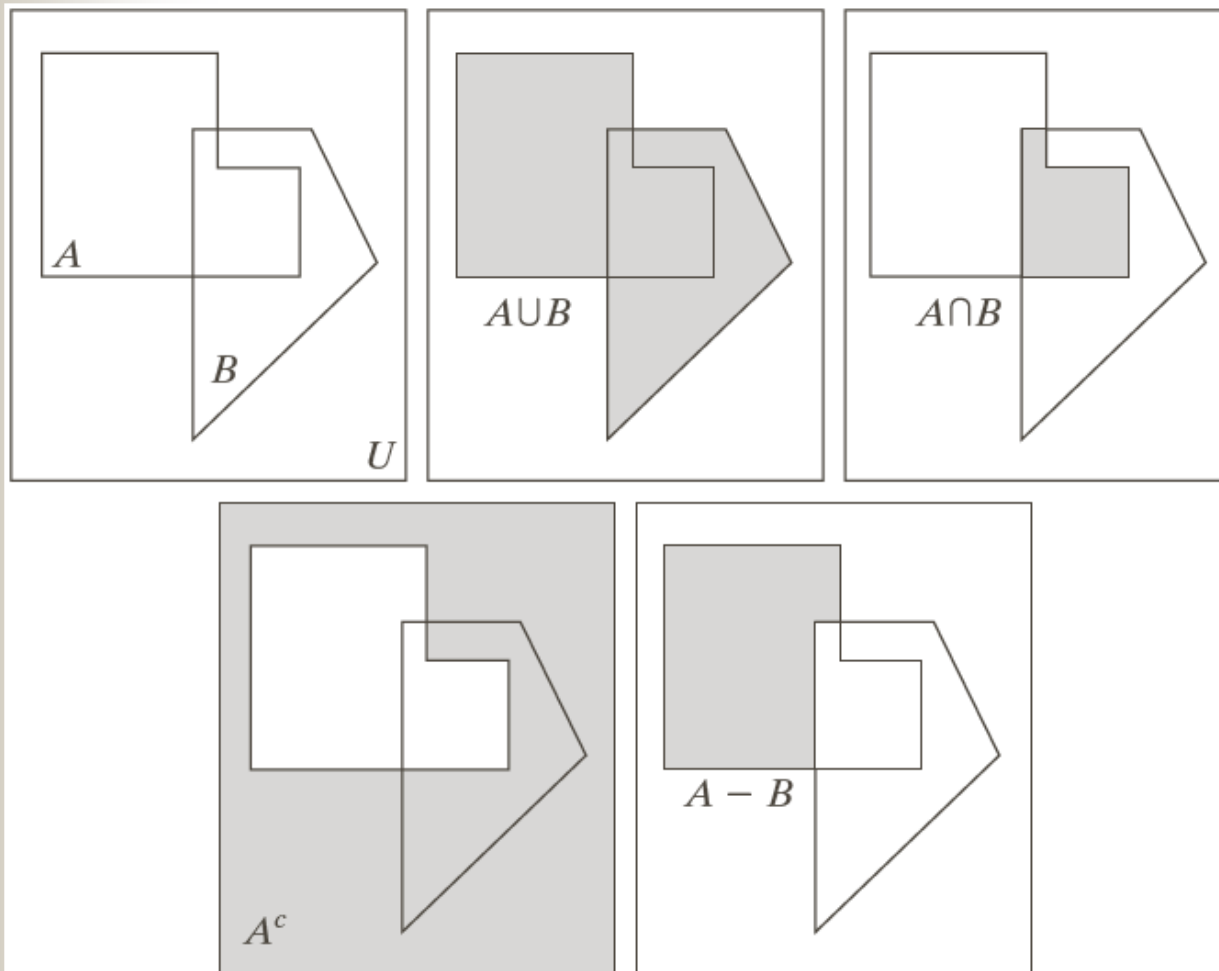
Netherlands.)



Image Arithmetic

- Addition
- Subtraction
- Multiplication
- Division
- Logical: AND, OR, XOR, NOT,...

Basic operations

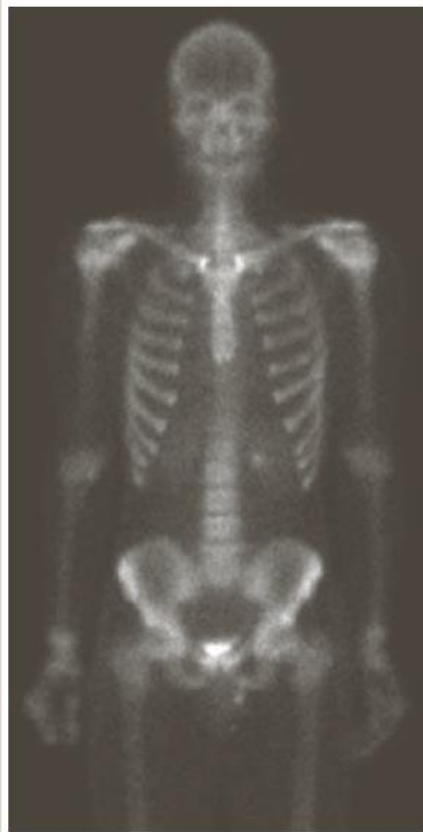


a	b	c
d	e	

FIGURE 2.31

(a) Two sets of coordinates, A and B , in 2-D space. (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B . In (b)–(e) the shaded areas represent the member of the set operation indicated.

Example of basic operations



a b c

FIGURE 2.32 Set operations involving gray-scale images. (a) Original image. (b) Image negative obtained using set complementation. (c) The union of (a) and a constant image. (Original image courtesy of G.E. Medical Systems.)

Logical Operations

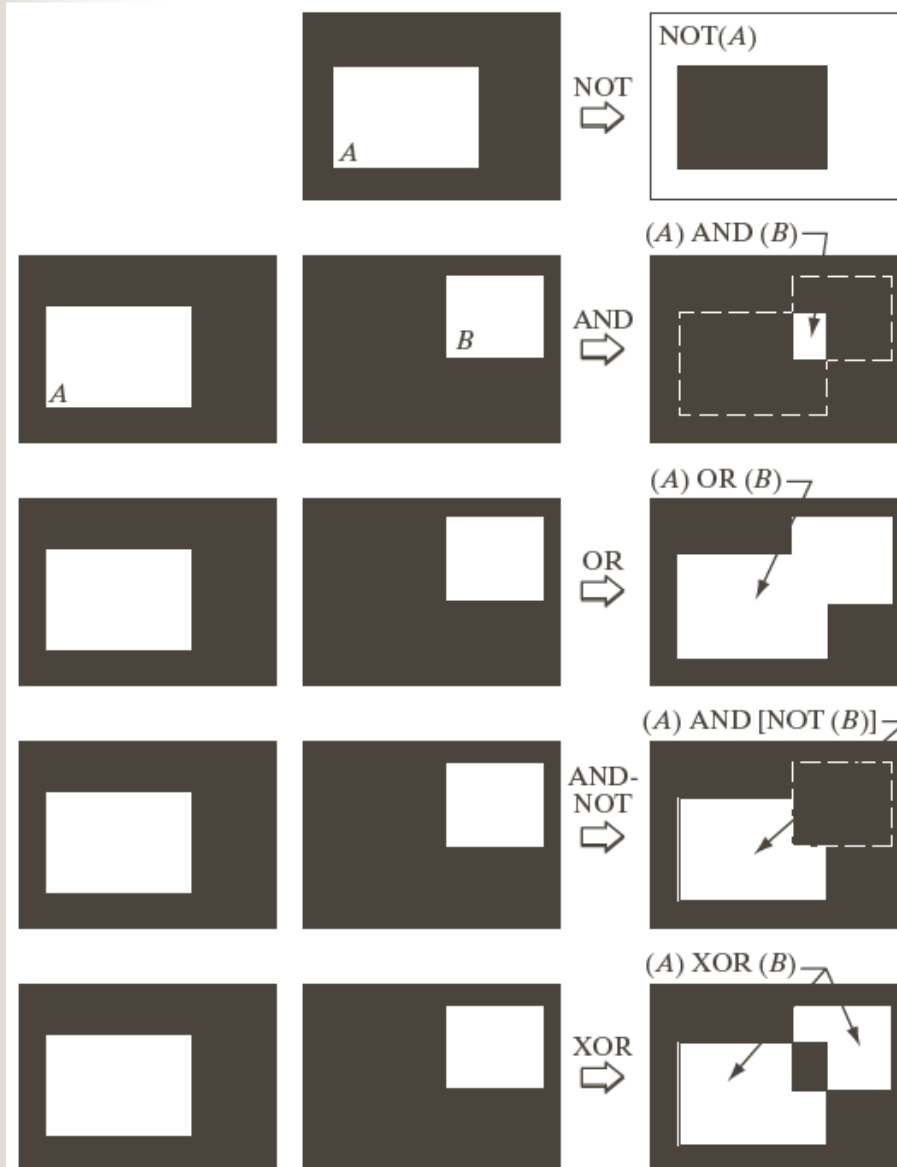


FIGURE 2.33

Illustration of logical operations involving foreground (white) pixels. Black represents binary 0s and white binary 1s. The dashed lines are shown for reference only. They are not part of the result.

Affine Transformation

TABLE 2.2

Affine transformations based on Eq. (2.6.–23).

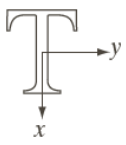
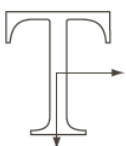
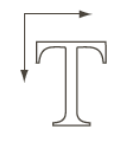
Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \\ y &= w \end{aligned}$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= c_x v \\ y &= c_y w \end{aligned}$	
Rotation	$\begin{bmatrix} \cos \theta & \sin \theta & 0 \\ -\sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \cos \theta - w \sin \theta \\ y &= v \sin \theta + w \cos \theta \end{aligned}$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$\begin{aligned} x &= v + t_x \\ y &= w + t_y \end{aligned}$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v + s_v w \\ y &= w \end{aligned}$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$\begin{aligned} x &= v \\ y &= s_h v + w \end{aligned}$	

Image Rotation



a b c d

FIGURE 2.36 (a) A 300 dpi image of the letter T. (b) Image rotated 21° clockwise using nearest neighbor interpolation to assign intensity values to the spatially transformed pixels. (c) Image rotated 21° using bilinear interpolation. (d) Image rotated 21° using bicubic interpolation. The enlarged sections show edge detail for the three interpolation approaches.

Image registration

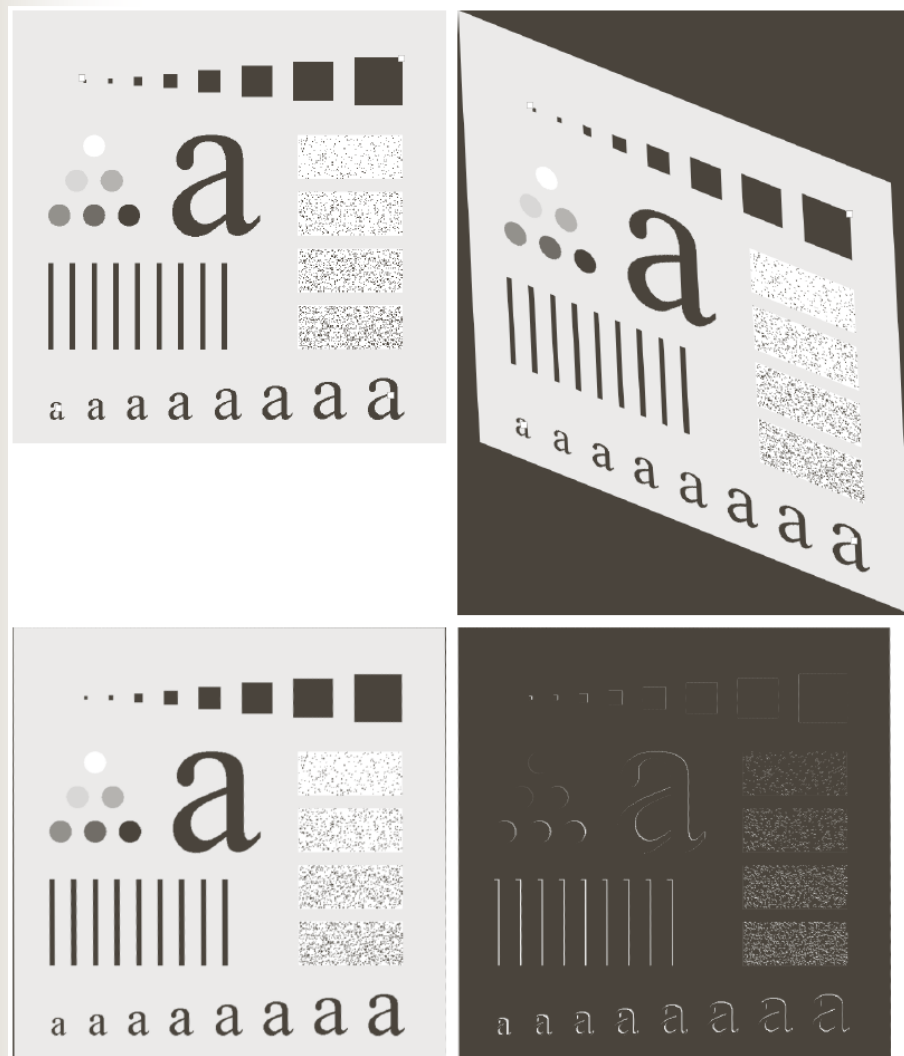


FIGURE 2.37

Image registration.
 (a) Reference image. (b) Input (geometrically distorted image). Corresponding tie points are shown as small white squares near the corners.
 (c) Registered image (note the errors in the borders).
 (d) Difference between (a) and (c), showing more registration errors.



HW 1 -- Due: 2/17/2026

- Discussions are welcomed.
- Never copy others' work (both students will receive no credit for copied homework).



Homework Requirements

- A word file with:
 - Questions
 - Your solutions with output images
- M files (matlab) -- executable
- Email the word file and matlab files to our TA:
 - Mr. Ryan Goldberg at
 - rgoldbe000@citymail.cuny.edu
 - Subject: I2200/EE4710 HW



Homework - Presentation

The student will present the HW and discuss with classmates.

- HW1 – Roy
- HW2 – Ali
- HW3 – Muzna

A good opportunity to practice presentation skills + **5 extra bonus points for that HW.**